# AQUABOTS Aeropak

## Integrated Status Review

Team 26061



# Overview

1.0 Introduction

2.0 Post-CDR Changes

3.0 Parts Status

4.0 Assembly and Integration

5.0 Testing and Verification

6.0 Risk Analysis

7.0 Budget and Deliverables

8.0 Summary

# Introduction

# Our Team

- Jamie Ledbetter, Biosystems Engineering – Team Lead
- Robert Wilkie, Aerospace Engineering – Procurement Lead
- Alex Imrich, Aerospace Engineering
- John Lopez, Electrical Engineering
- Mayra Mendoza, Optical Engineering
- Kama Henriques, Mechanical Engineering

# Project Summary

**Purpose:** Rapid, real-time environmental monitoring in marine and coastal environments

**Deployment:** Ocean-deployed drones self-orient within 1 minute and follow remotely set paths

**Ground Station:**

- Live video stream from main drone
- Real-time environmental data from all drones
- GEO-overlay for spatial monitoring and mission control

**Hull & Structure:**

- Self-righting, waterproof, impact-resistant design
- Fiberglass enclosure protecting internal systems

**Communication:**

- Starlink satellite antenna for continuous remote connectivity

**Onboard Systems:**

- Jetson Nano – Controls motors, servos, and live video
- ESP32 – Collects GPS, time, altitude, temperature, pH, and conductivity data



*Image of AQUABOT Hull*

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# Post-CDR Changes

# Post-CDR Changes

- **Action Items: Detailed integration plans for both software and hardware**
- Modified sensor selection– prior pH sensors faulty due to improper maintenance
- Modified sensor and camera mounts to provide more stability and minimize drag
- Added flight controller to reduce control software risks

All Action Items Closed

# Parts Status

# Communications

| Part Name | Weight | Supplier | Status |
|---|---|---|---|
| ESP32 | 1g | John | Purchased |
| Jetson Orin Nano | 0.5 kg | Amazon | Purchased |
| Starlink Antenna | 2 kg | Prior Team | On Hand |
| JST-SH 4-pin | - | Amazon | Purchased |
| Servo 3-pin | - | Amazon | Purchased |
| 15-pin FPC | - | Amazon | Purchased |
| RJ-45/Cat 5E Cable | 0.5 kg | Amazon | Purchased |

# Electronics

All parts ordered • Awaiting delivery

| Part | Qty | Price | Source | Status |
|---|---|---|---|---|
| 6S LiPo Battery (main) | 1 | $66 | Amazon | ORDERED |
| MRBF Fuse Holder (main) | 1 | $25 | Amazon | ORDERED |
| MIDI Fuse Block (thruster) | 1 | $38 | Amazon | ORDERED |
| ST Blade Fuse Block (aux) | 1 | $47 | Amazon | ORDERED |
| 150A Fuse (main) | 1 | $18 | Amazon | ORDERED |
| 125A Fuse (motor) | 1 | $12 | Amazon | ORDERED |
| Bus Bars (+ / –) | 2 | $100 | Amazon | ORDERED |
| Manual Kill Switch | 1 | $25 | Amazon | ORDERED |
| Castle BEC (servo 7.4V) | 1 | $47 | Castle | ORDERED |
| Thermal Pads + ATO Fuses + DC Pigtails | — | $39 | Amazon | ORDERED |

**Total: $417**



All parts ordered. Awaiting delivery. Assembly begins upon arrival — no design work remaining, straight to build.

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# Propulsion System/Mechanical

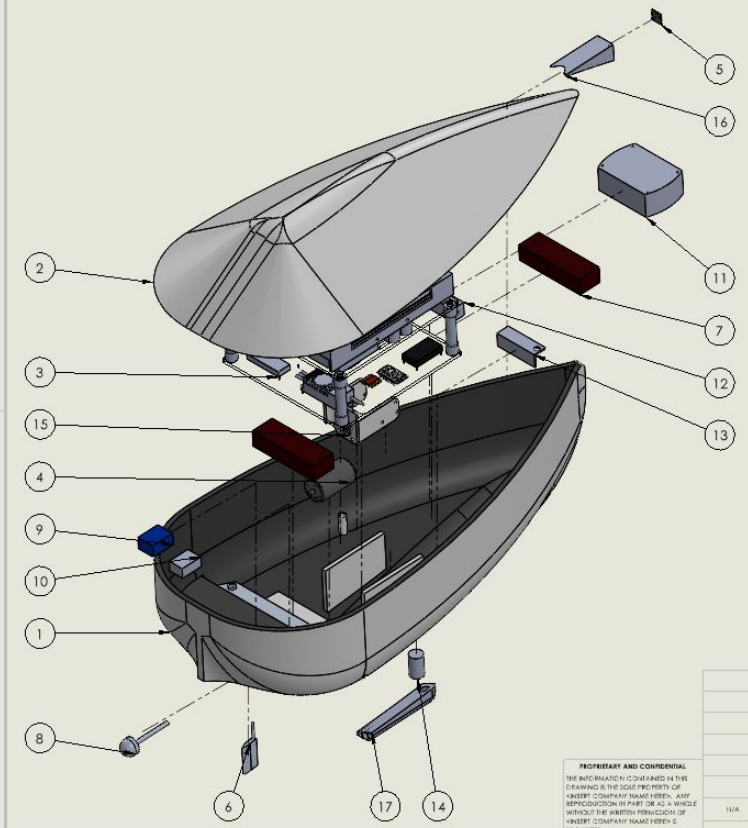
| Part Name | Weight | Supplier | Status |
|---|---|---|---|
| Motor | 680g | Apsiqueen | Installed |
| Rudder Servo | 90g | DSServo | Shipping |
| Shaft | 150g | | Installed |
| Propeller | 95g | 3D Printed (3DP) | Installed |
| Battery | 825g | OVONIC | Ordering |
| Motor Mount | 82g | 3DP | Installed |
| Motor Flange | ~100g | 3DP/ Machine Shopped | Under Redesign |
| PIXHAWK 4X | 30g | PIXHAWK | Will install once servo arrives |
| ESC | 120g | Apsiqueen | Installed |
| Rudder | 30g | Machine Shopped | Installed |

# Sensors and Camera

| Part Name | Weight | Supplier | Status |
|---|---|---|---|
| IMX219 | 10 g | Amazon | Purchased, Installed |
| PINPOINT Marine pH Probe | 29 g | Saltwater Aquarium | Awaiting Arrival |
| EZO pH Circuit | 1.76 g | Atlas Scientific | Purchased |
| DS18B20 Waterproof Temp. Sensor | 25 g | Amazon | Awaiting Install |
| TDS Conductivity Sensor | 27.2 g | Amazon | Awaiting Install |
| Sensor Mount | TBD | CAD Part | Pending Final Print |
| Camera Mount | TBD | CAD Part | Printed, Installed |

# Assembly and Integration Status

# AQUABOT Assembly Drawing




| ITEM NO. | PART NUMBER | DESCRIPTION | QTY. |
|---|---|---|---|
| 1 | AQ401_BottomShell-1 | Lower Shell | 1 |
| 2 | AQ402_TopShell-1 | Upper Shell | 1 |
| 3 | AQ202_ShelfAssem-1 | Shelf Assembly | 1 |
| 4 | AQ418_MainMotor-1 | Motor for Prop | 1 |
| 5 | AQ423_Camera-1 | Front-facing Camera | 1 |
| 6 | AQ428_Rudder-1 | Rudder for Steering | 1 |
| 7 | AQ416-LiPO-1 | Battery | 2 |
| 8 | AQ434_PropellerScaled-1 | Propeller | 1 |
| 9 | AQ435_Servo-1 | Rudder Control Servo | 1 |
| 10 | AQ432_BEC-1 | BEC | 1 |
| 11 | AQ436_KillSwitch-1 | Physical Kill Switch | 1 |
| 12 | AQ441_LeftWingFlange-1 | Left Shelf-support | 1 |
| 13 | AQ442_RightWingFlange-1 | Right Shelf-support | 1 |
| 14 | AQ443_TempSensor-1 | Temperature Sensor | 1 |
| 15 | AQ420_MotorMount-2 | Motor Mount | 1 |
| 16 | AQ419_CameraHousing-2 | Flush Camera Housing | 1 |
| 17 | AQ425_SensorMount2023-2 | External Sensor Mount | 1 |

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UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:
FRACTIONAL ±
ANGULAR: MACH ± BEND ±
TWO PLACE DECIMAL ±
THREE PLACE DECIMAL ±
INTERPRET GEOMETRIC TOLERANCING PER:
MATERIAL
FINISH
NEXT ASSY
USED ON
N/A
APPLICATION
DO NOT SCALE DRAWING

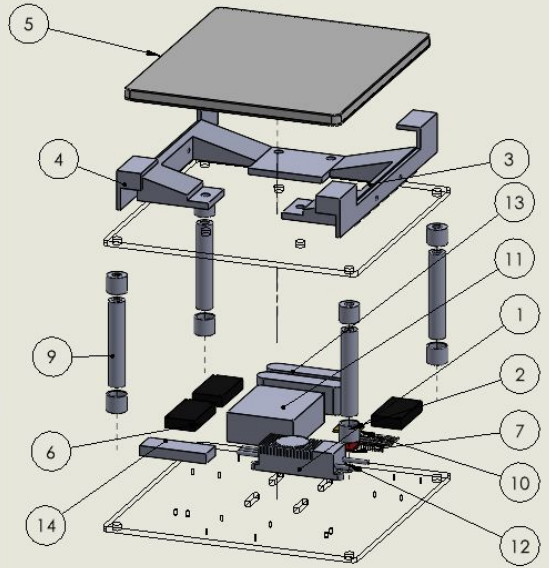
| | NAME | DATE |
|---|---|---|
| DRAWN | | |
| CHECKED | | |
| ENG APPR. | | |
| MFG APPR. | | |
| Q.A. | | |
| COMMENTS: | | |

TITLE: AQUABOT Assembly

| SIZE | DWG. NO. | REV |
|---|---|---|
| B | D201-1 | 2 |

SCALE: 1:6 WEIGHT: SHEET 1 OF 1

# Shelf Subassembly Drawing




| ITEM NO. | PART NUMBER | DESCRIPTION | QTY. |
|---|---|---|---|
| 1 | AQ403_BottomShelf-1 | Bottom Shelf | 1 |
| 2 | AQ409_GPS-1 | Gloal Positioning System | 1 |
| 3 | AQ404_TopShelf-1 | Top Shelf | 1 |
| 4 | AQ203_StarlinkMount-1 | Starlink Mounting System | 1 |
| 5 | AQ407_Starlink-1 | StarLink Mini | 1 |
| 6 | AQ412_StepDownConverter-1 | Step Down Converter | 3 |
| 7 | AQ410_IMU-1 | Inertial Measurement Unit | 1 |
| 8 | AQ413_ShelfSupportEndCaps-1 | Support End Caps | 8 |
| 9 | AQ408_SupportColumns-1 | Support Columns | 4 |
| 10 | AQ429_2023Esp32-1 | Esp32 Module | 1 |
| 11 | AQ417_JetsonNano-1 | Jetson Nano | 1 |
| 12 | AQ426_ESC-1 | ESC | 1 |
| 13 | AQ430_BusBar-1 | Bus Bar | 2 |
| 14 | AQ437_EStopRelay-1 | E-Stop Relay | 1 |

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MM
TOLERANCES:
FRACTIONAL ±
ANGULAR: MACH ± BEND ±
TWO PLACE DECIMAL ±
THREE PLACE DECIMAL ±
INTERPRET GEOMETRIC TOLERANCING PER:
MATERIAL
FINISH
DO NOT SCALE DRAWING

| | NAME | DATE |
|---|---|---|
| DRAWN | | |
| CHECKED | | |
| ENG APPR. | | |
| MFG APPR. | | |
| Q.A. | | |

COMMENTS:

APPLICATION: 201 Aquabot
USED ON:

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TITLE: Shelf Assembly

SIZE A | DWG. NO. D202-1 | REV

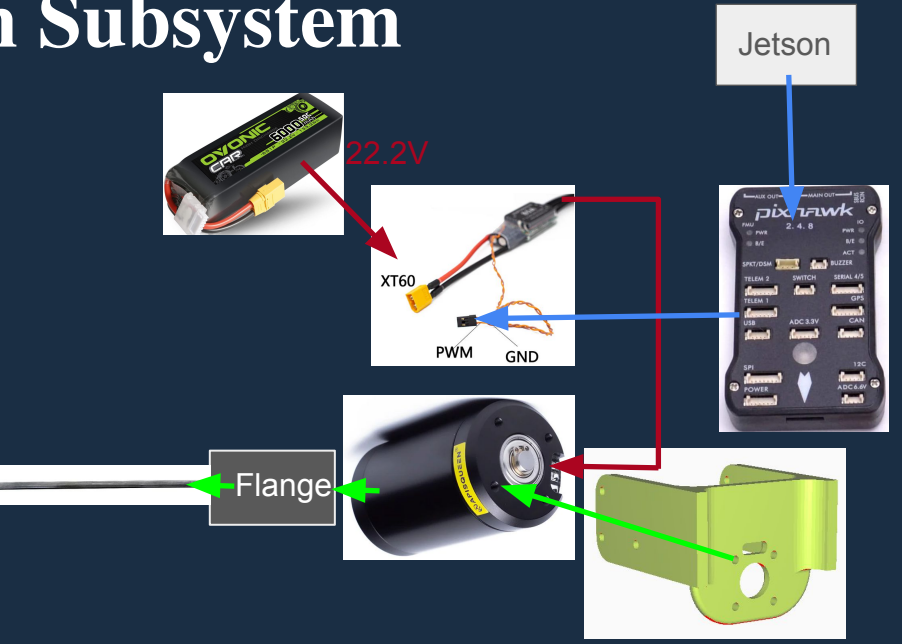
SCALE: 1:10 WEIGHT: | SHEET 1 OF 1

# Propulsion Subsystem

Data Connection

Power Connection

Physical Connection



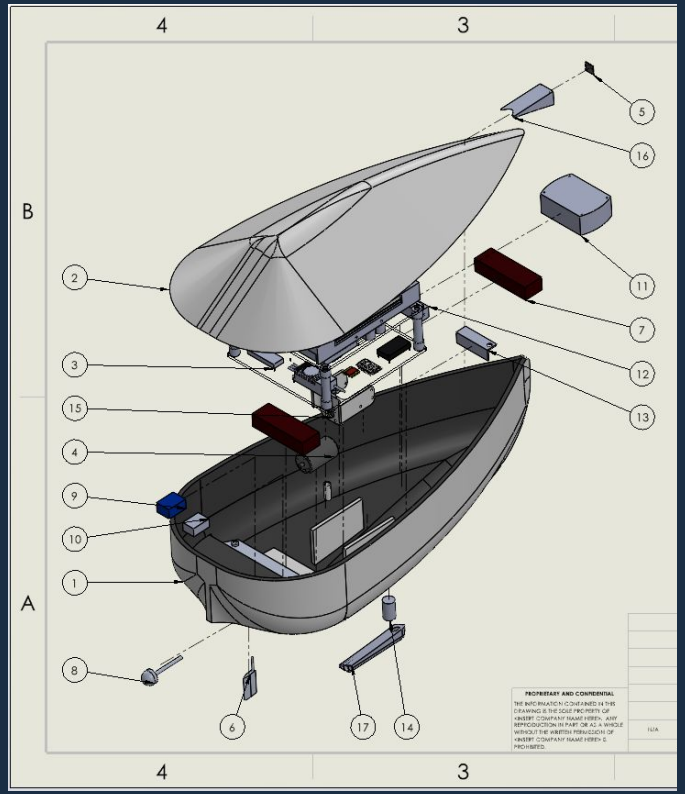



Integration Update:

- Assembled Motor, mount, shaft, and propeller
- Waiting on rudder servo for full assembly
- Testing complete: waterproof shaft assy, motor rotation in hull

# Propulsion Subsystem Integration Plan

1. Motor Mounting - Complete by 3/2/2026
   a. Ensure motor mount holes are aligned with holes on inner-drone mount
   b. Install and attach 3D Printed Motor Mount Using 4 ¼” bolts and nuts
   c. Place motor into mount, threading wires toward electronics stack
   d. Line motor up with propeller shaft (already in place)
   e. Use joint connection to attach propeller shaft to motor
   f. Attach motor to power and computer, verify functionality after install
2. Servo Mounting - Complete by 3/7/2026
   a. Assess whether servo adhesive is stable enough, otherwise a 3D-printed mount may be needed.
   b. Adhere servo 1 inch behind rudder shaft
   c. Use wire and joint connector to connect servo to rudder shaft
   d. Attach servo to power and computer, verify functionality after install
3. Electronics Stack - Complete by 3/15/2026
   a. Reintegrate 1st level of stack into hull, as placed before
   b. Ensure all connections are properly soldered before installation into stack
   c. Use velcro to place electronics as outlined by software int. document
   d. Secure and organize all wires, ensure power supplies supported
   e. Add 2nd level to stack – with Starlink Antenna Mount




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# Self-Righting and Waterproofing Subsystem Testing/Verification (5.1.3, 5.3.2)

| Analysis of Torque due to Mass Offset | | | | | |
|---|---|---|---|---|---|
| | Approximate Center of Buoyancy (mm) | Assembly Center of Mass (mm) | Offset (mm) | Total Mass (kg) | Resulting Torque (N*m) |
| X (Roll) | 0 | 1.23 | -1.23 | 29.26 | -0.353 |
| Y (Yaw) | -326.54 | -296.47 | -30.07 | | 0.000 |
| Z (Pitch) | -303.89 | -281.52 | -22.37 | | -6.421 |

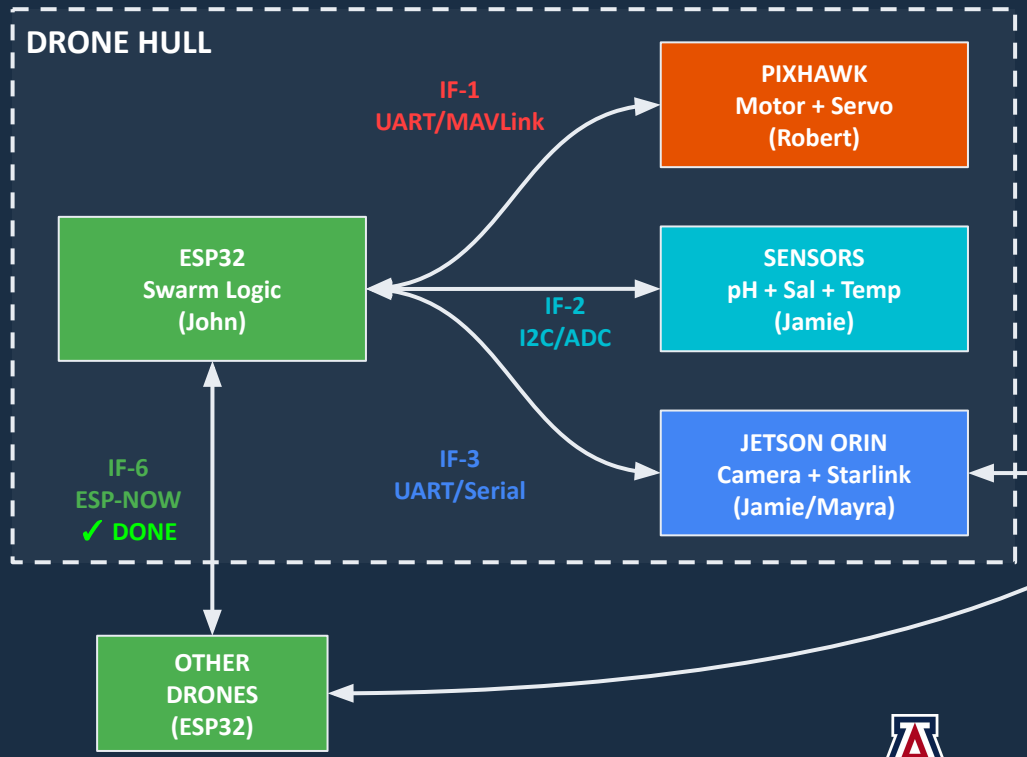


*Image showing initial waterproof and self-righting testing*

- COM offset only results in a small change in equilibrium pitch
- Calculated low COM indicates self-righting capabilities after modifications
- Initial self-righting and waterproof testing successful
- Extra waterproofing precautions: possible bilge pump implementation, upgraded washers, waterproof sealant
- Final testing will be conducted before 3/30/2026

# Communications System — Block Diagram

6 integration interfaces across 4 subsystems




## Interface Status

- IF-1: Swarm↔Pixhawk — Spec Wk 8
- IF-2: Sensors→ESP32 — Spec Wk 8
- IF-3: ESP32↔Jetson — Spec Wk 8
- IF-4: Drone↔Cloud — Spec Wk 8
- IF-5: Cloud↔GS — Spec Wk 8
- IF-6: Drone↔Drone — ✓ DONE

# 9-Step Development Roadmap

Incremental build — each step tested before advancing

**1 Scaffolding**
✓ COMPLETE
20-file modular arch
ESP-NOW wrapper

**2 Identity**
✓ COMPLETE
NVS provisioning
Persistent node IDs

**3 Discovery**
✓ COMPLETE
HELLO/JOIN protocol
50-node capacity

**4 Election**
✓ COMPLETE
Leader + TDMA
50ms slots/beacons

**5 Commands**
✓ COMPLETE
GOTO/HOLD/STOP
Synced execution

**6 Formation**
✓ COMPLETE
4 patterns + move
Leader-distributed

**7 Sensor Scaffold**
PLANNED
Sensor abstraction
Mock data API

**8 Ultrasonic Distance Sensor**
PLANNED
Real sensor driver
Object detection

**9 Avoidance**
PLANNED
Collision avoid
Path planning

# Swarm Communications — Status

ESP32-S3 / ESP-NOW • 6 of 9 development steps complete

## Development Progress: 6 / 9 Steps

67%

| 6/9 | 50 | 0% | <5ms | <50ms |
|---|---|---|---|---|
| Steps Complete | Max Swarm Size | Packet Loss | RTT Latency | Command Latency |

### Completed (Steps 1-6)

- Step 1: Modular scaffolding + transport
- Step 2: NVS persistent node identity
- Step 3: HELLO/JOIN auto-discovery (2-4 sec)
- Step 4: Leader election + TDMA scheduling
- Step 5: Command engine (GOTO/HOLD/STOP/RESUME)
- Step 6: Formation control (line/wedge/circle/individual)

### Remaining (Steps 7-9)

- Step 7: Obstacle sensor scaffolding
- Step 8: mmWave radar integration
- Step 9: Avoidance policy tuning
- **FINAL STEP: Debugging and Stress Testing**

# Formation Control — Verified

Leader-distributed • Extensible pattern API • Thread-safe

## Built-in Formation Patterns

LINE CHEVRON CIRCLE INDIVIDUAL



## How It Works

- Leader sets formation type + spacing
- Config broadcast to all nodes via ESP-NOW
- Each node calculates its own position from index
- Extensible pattern system (function pointers)

## Verified on Hardware — Feb 09, 2026

| Test | Leader (idx 0) | Follower (idx 1) | Result |
|---|---|---|---|
| LINE 2000mm | (0, 0) | (0, -2000) | ✓ PASS |
| WEDGE 2000mm | (0, 0) | (-1000, -2000) | ✓ PASS |
| WEDGE 500mm | (0, 0) | (-250, -500) | ✓ PASS |
| Spacing change | — | Scaled correctly | ✓ PASS |
| Non-leader | — | Rejected | ✓ PASS |

## Console Commands

```
formation <type> • spacing <mm> • fmove <x> <y> • fstatus
```

## All Previous Steps Still Working

Discovery ✓ | Election ✓ | TDMA ✓ | GOTO ✓ | HOLD ✓

STOP ✓ | RESUME ✓ | Failover ✓ | Thread Safety ✓

# Hardware & Verified Metrics

ESP32-S3 • All performance metrics exceed requirements

## ESP32-S3 Platform

- MCU: Xtensa Dual-Core @ 240 MHz
- Radio: 2.4 GHz WiFi (ESP-NOW protocol)
- Range: 30m+ indoor, 100m+ outdoor (line of sight)
- UART0=Console, UART1=Pixhawk, UART2=Jetson
- RTOS: FreeRTOS • Storage: NVS Flash
- Currently verified on 2 dev boards

## Resource Usage (per node)

| Resource | Usage |
|---|---|
| Flash | ~800 KB |
| RAM | ~10 KB |
| CPU Load | <5% |
| Network Traffic | ~50-100 B/sec |
| Modules | 9 (6 active, 3 stub) |

**4-5 ms**

RTT Latency

Req: <10 ms

PASS

**0%**

Packet Loss

Req: <5%

PASS

**-16 to -24 dBm**

Signal (RSSI)

Req: >-70 dBm

PASS

**2-4 sec**

Discovery Time

Req: <5 sec

PASS

# SRVM Verification – Requirement 5.4.2

Inter-Drone Communication: PARTIAL PASS ⟶ Advancing to FULL PASS

| Verification Method | Status | Evidence / Detail | Date |
|---|---|---|---|
| TEST | ✓ PASS | RTT <5ms \| 0% loss \| RSSI -16 to -24 dBm \| Discovery <5s | Wk 6 |
| DEMONSTRATION | ✓ PASS | 2-node swarm: auto-discovery, leader election, TDMA, cmd sync | Wk 6 |
| ANALYSIS | ◐ In Progress | Scalability 2⟶50 nodes, traffic modeling, performance projections | Wk 9 |
| INSPECTION | ○ Planned | Hardware inspection, antenna placement, enclosure RF check | Wk 12 |

## Other Requirements

- 5.3.1 Survivability – ◐ Test: ✓ COMMS PASS (0% packet loss w/ calm conditions) | Analysis: Scheduled Wk 9
- 5.4.3 Data Relay – Demonstration: ◐ PARTIAL (positional data sent, sensor data pending)
- 5.4.1 Autonomy & Control – Demonstration: ◐ PARTIAL (command execution OK, formation pending) - NO CHANGE

# Power Subsystem/Electrical

Design unchanged from CDR • Fuse protection upgraded • All parts ordered

## Changes Since CDR

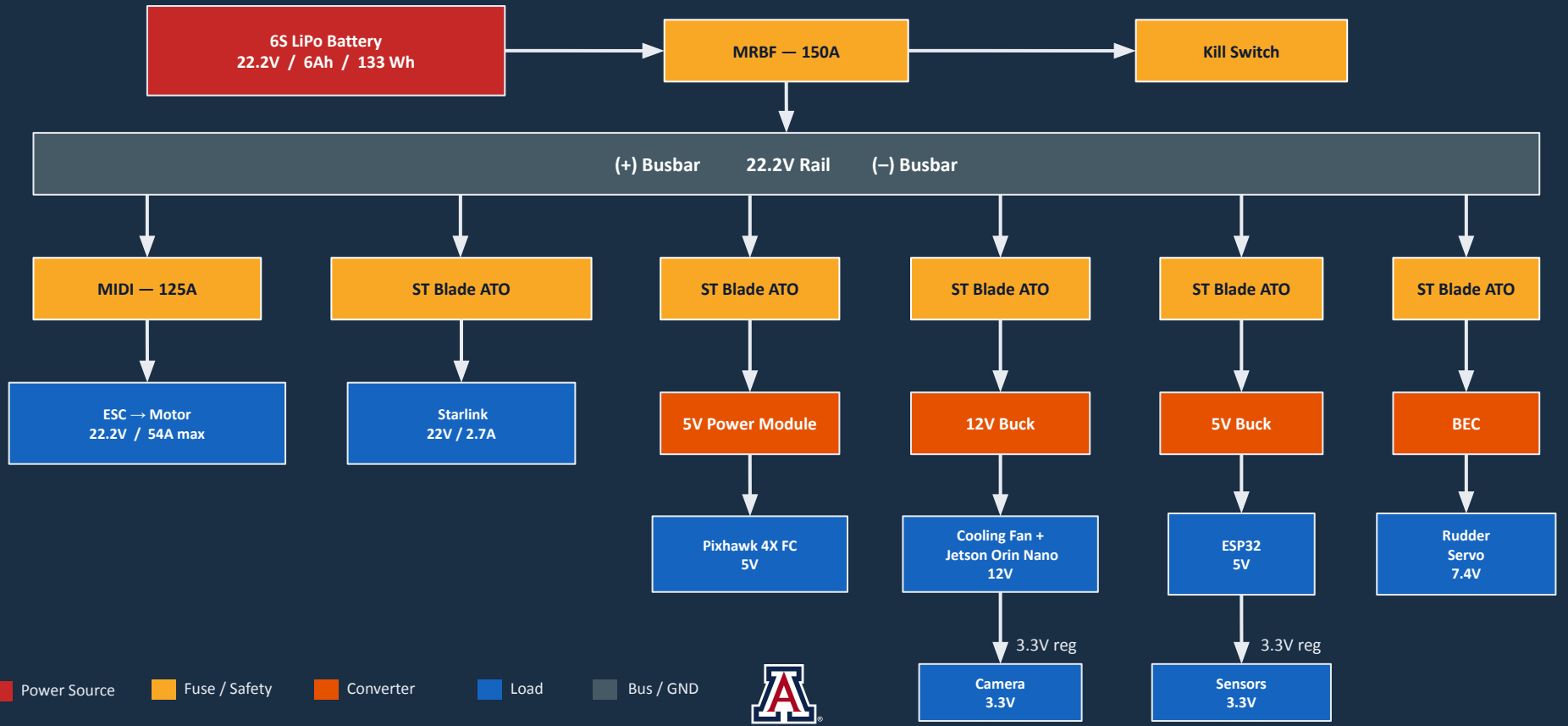
- Main fuse upgraded: 80A → 150A (MRBF holder)
- Added dedicated 125A motor fuse (MIDI block)
- Proper fuse blocks: MRBF + MIDI + ST Blade ATO
- Added thermal pads for heat management

## Unchanged from CDR

- 6S LiPo battery architecture (22.2V)
- Busbar distribution (+/- rails)
- Voltage converters: 5V, 12V buck
- Castle BEC for rudder servo (7.4V)
- Manual kill switch + master safety

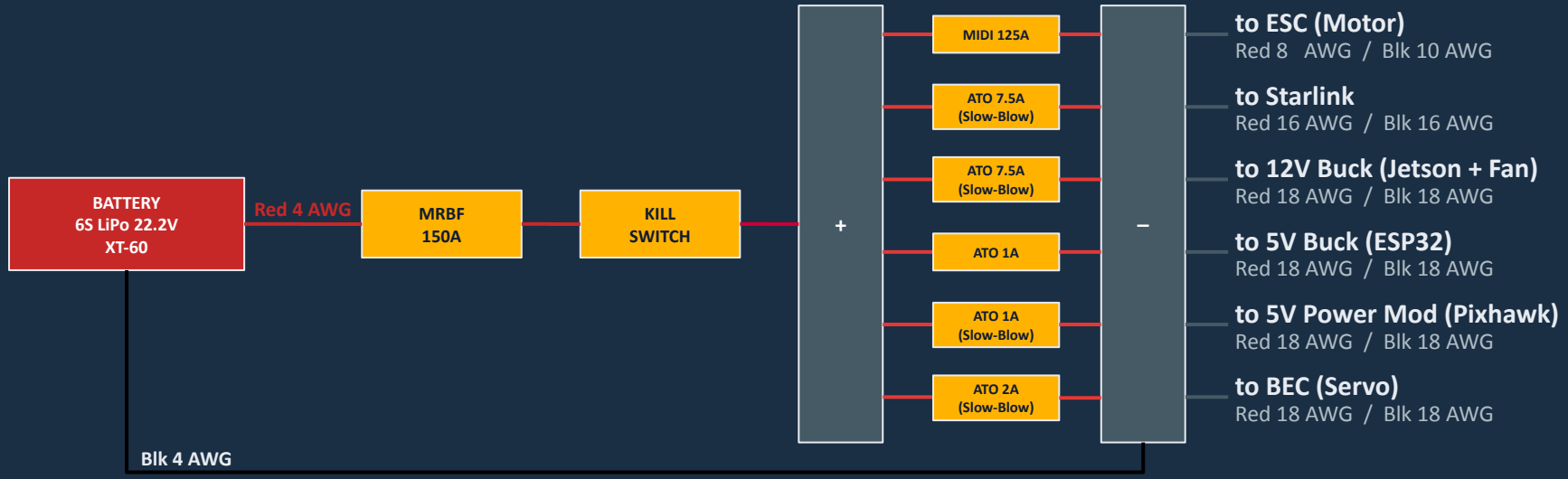
| 22.2V | 150A | 125A | ~155W | ~51 min |
|---|---|---|---|---|
| Battery Voltage | Main Fuse | Motor Fuse | Cruise Draw | Cruise Endurance |
| 6S LiPo | MRBF Block | MIDI Block | All Systems | 133.2 Wh |

# Power System — Block Diagram




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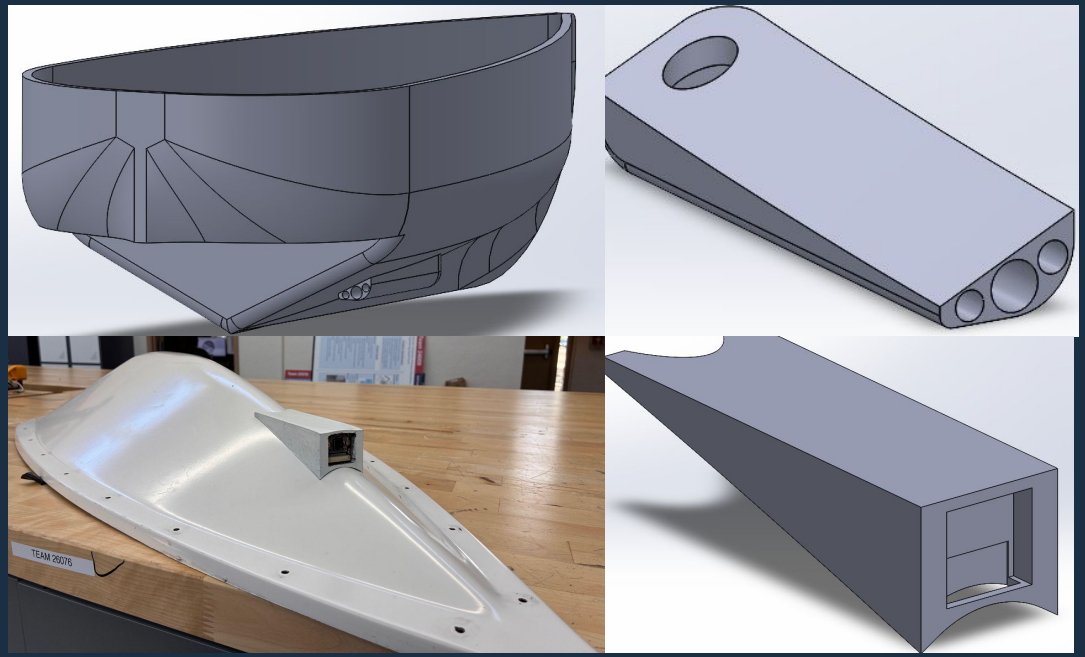
# Wiring Diagram — Interconnects & Cable Sizing




## Signal & Data Interconnects

| Connection | Protocol | Signal | Cable | Connector |
|---|---|---|---|---|
| ESP32 → Pixhawk (IF-1) | MAVLink / UART1 | TX/RX 3.3V | 22 AWG | JST-SH 4-pin |
| ESP32 → Jetson (IF-3) | UART2 115200 | TX/RX 3.3V | 22 AWG | JST-SH 4-pin |
| ESP32 → ESC (PWM) | PWM 50Hz | 3.3V logic | 22 AWG | Servo 3-pin |
| Jetson → Camera (CSI) | MIPI CSI-2 | LVDS diff | Ribbon | 15-pin FPC |
| Jetson → Starlink | Ethernet | 100BASE-T | Cat5e | RJ-45 |

# Sensors and Camera Integration




Progress Since CDR:

- Mounts optimized
- Drag reduced for sensor mount
- Stability improved on camera mount
- Camera successfully mounted
- New pH sensor selected – prior choice faulty CDE2 Delay

Next Steps:

- Waiting on pH sensor arrival
- Program pH sensor inline with temp. and conductivity sensor
- Integrate into hull

# Critical Design Element (CDE-2) Progress

CDE-2 Goal:

Demonstrate sensor and camera connectivity to the ground station GUI, while being integrated into the drone hull itself.

Current Status:

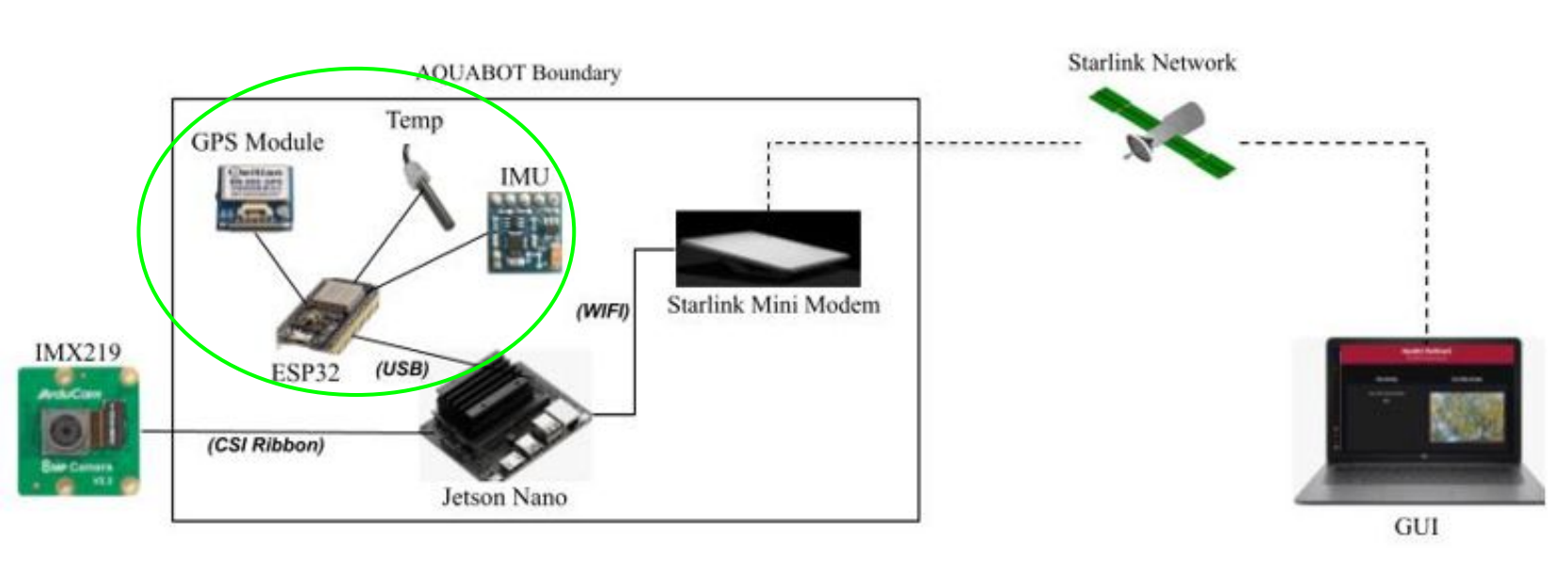
Camera is integrated with partial verification of functionality. Sensor suite still awaiting integration due to pH sensor replacement– functionality of temperature and conductivity sensor is partially verified.

Next Steps:

After pH sensor arrives, sensor suite programming will be finalized (1-2 days), and then mounted (1 day).

Verification of function within hull & waterproofing will be done at pool by 3/3/26.

# Geo-Overlay and Telemetry Integration




- IMX219 camera sends 1080p video to the Jetson via CSI ribbon
- ESP32 collects GPS, IMU, and temperature data and sends telemetry to Jetson over USB
- Jetson synchronizes video + telemetry and prepares outgoing data stream
- Jetson connects to Starlink Mini via Wifi for remote communication
- Starlink network relays live video + telemetry to the GUI
- GUI displays real-time video, GPS position, heading, and temperature and supports two-way commands

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# Demo Setup

Starlink Mini

IMX219



Power for Jetson & Starlink



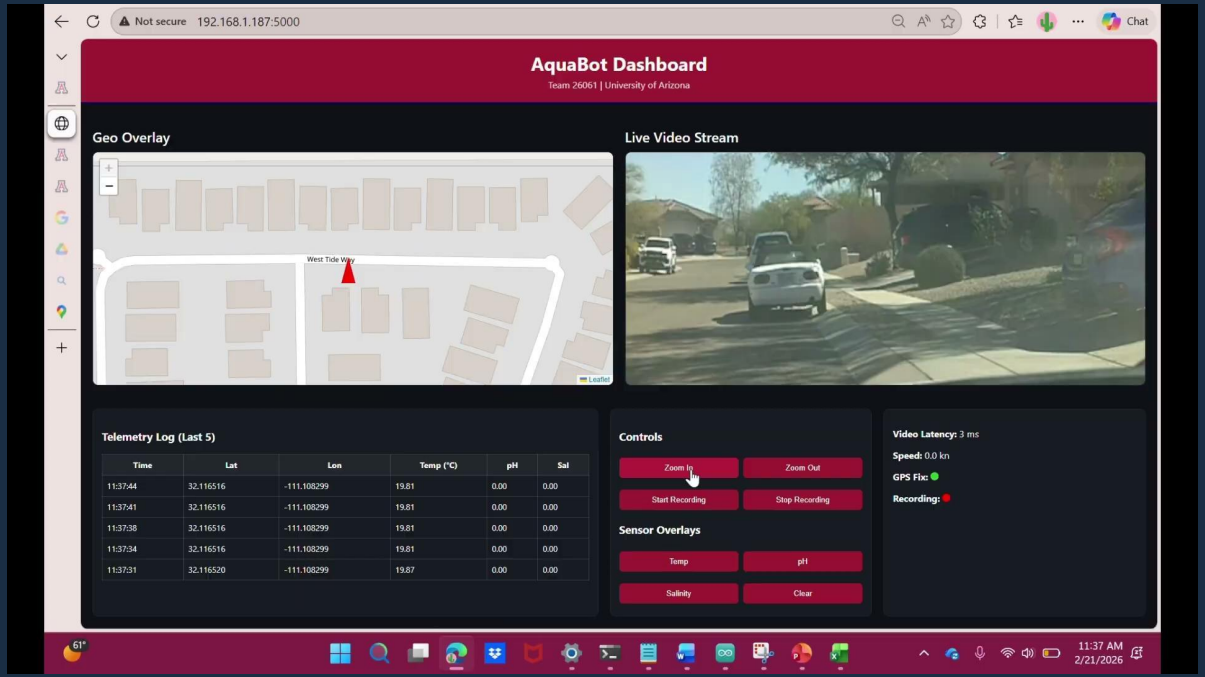
Starlink Network Connection




Jetson Nano

ESP32: GPS, IMU & Temp

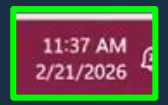
# Geo-Overlay Demo




## Results:

- Live 1080p video successfully streamed over Starlink
- Geo-overlay displayed correctly with real-time GPS updates
- Telemetry (GPS, IMU, Temp) received at expected 3-second interval
- GUI controls functional (zoom, record, overlays)
- Recording triggered from GUI and saved on Jetson with matching timestamps
- End-to-end Jetson ↔ GUI communication confirmed

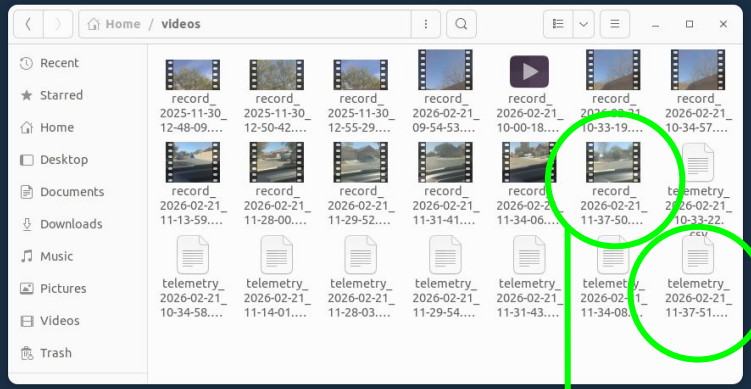
**For Reference:**

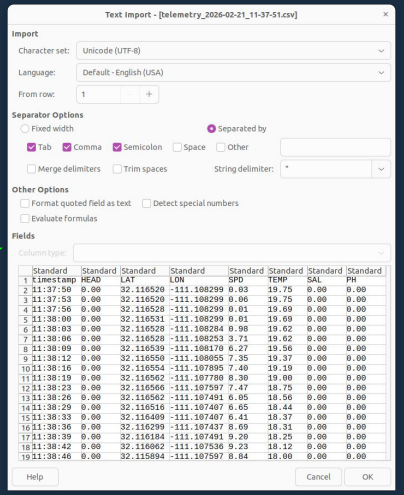



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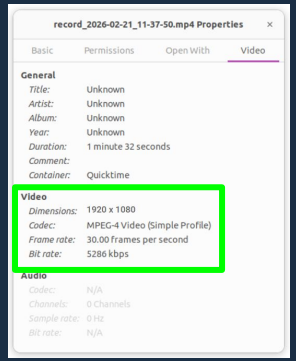
# Testing and Verification

# Data Transmission Verification




| | Standard | Standard | Standard | Standard | Standard | Standard | Standard | Standard |
|---|---|---|---|---|---|---|---|---|
| 1 | timestamp | HEAD | LAT | LON | SPD | TEMP | SAL | PH |
| 2 | 11:37:50 | 0.00 | 32.116520 | -111.108299 | 0.03 | 19.75 | 0.00 | 0.00 |
| 3 | 11:37:53 | 0.00 | 32.116520 | -111.108299 | 0.06 | 19.75 | 0.00 | 0.00 |
| 4 | 11:37:56 | 0.00 | 32.116520 | -111.108299 | 0.01 | 19.69 | 0.00 | 0.00 |
| 5 | 11:38:00 | 0.00 | 32.116531 | -111.108299 | 0.01 | 19.69 | 0.00 | 0.00 |
| 6 | 11:38:03 | 0.00 | 32.116528 | -111.108284 | 0.98 | 19.62 | 0.00 | 0.00 |
| 7 | 11:38:06 | 0.00 | 32.116528 | -111.108253 | 3.71 | 19.62 | 0.00 | 0.00 |
| 8 | 11:38:09 | 0.00 | 32.116539 | -111.108170 | 6.27 | 19.56 | 0.00 | 0.00 |
| 9 | 11:38:12 | 0.00 | 32.116550 | -111.108055 | 7.35 | 19.37 | 0.00 | 0.00 |
| 10 | 11:38:16 | 0.00 | 32.116554 | -111.107895 | 7.40 | 19.19 | 0.00 | 0.00 |
| 11 | 11:38:19 | 0.00 | 32.116562 | -111.107780 | 8.30 | 19.00 | 0.00 | 0.00 |
| 12 | 11:38:23 | 0.00 | 32.116566 | -111.107597 | 7.47 | 18.75 | 0.00 | 0.00 |
| 13 | 11:38:26 | 0.00 | 32.116562 | -111.107491 | 6.05 | 18.56 | 0.00 | 0.00 |
| 14 | 11:38:29 | 0.00 | 32.116516 | -111.107407 | 6.65 | 18.44 | 0.00 | 0.00 |
| 15 | 11:38:33 | 0.00 | 32.116409 | -111.107407 | 6.41 | 18.37 | 0.00 | 0.00 |
| 16 | 11:38:36 | 0.00 | 32.116299 | -111.107437 | 8.69 | 18.31 | 0.00 | 0.00 |
| 17 | 11:38:39 | 0.00 | 32.116184 | -111.107491 | 9.20 | 18.25 | 0.00 | 0.00 |
| 18 | 11:38:42 | 0.00 | 32.116062 | -111.107536 | 9.23 | 18.12 | 0.00 | 0.00 |
| 19 | 11:38:46 | 0.00 | 32.115894 | -111.107597 | 8.84 | 18.00 | 0.00 | 0.00 |







## 4.1.6 Data Collection

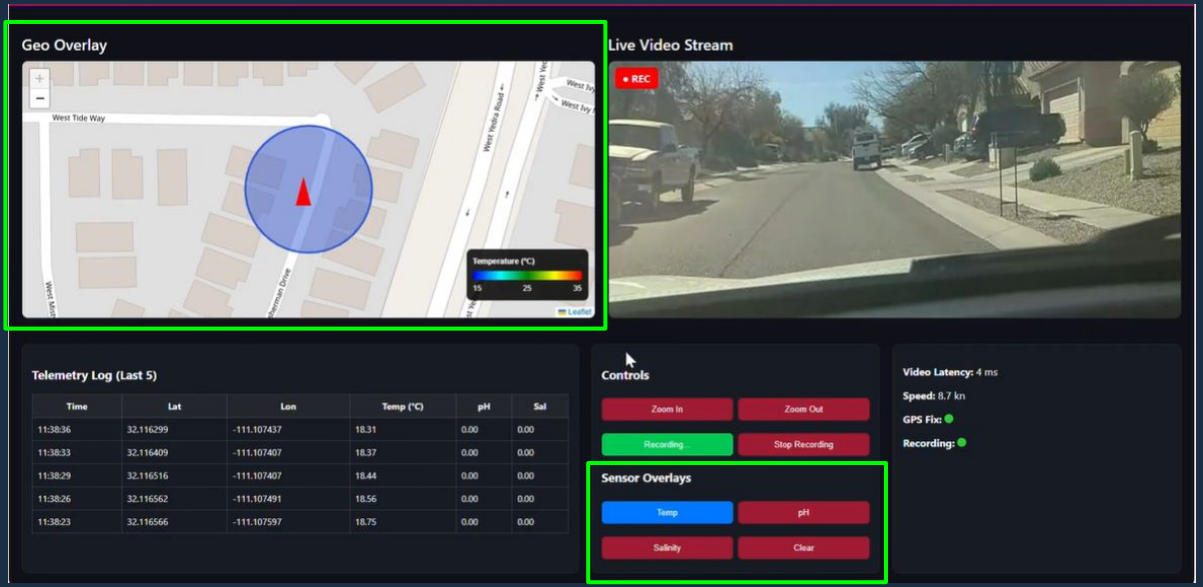
- GUI triggers recording on Jetson
- 1080p/30fps video verified
- Telemetry logged every 3 seconds
- Video + CSV stored with matching timestamps
- Jetson ↔ GUI communication confirmed

## 4.1.7 Data Mapping

- GPS + Temp exported as GIS-compatible CSV
- Synchronized video + telemetry
- Real-world mapping validated

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# Geo-Overlay Verification




- Map updates with GPS
- Temperature overlay toggles correctly
- Real-time position and heading displayed
- Live-Video Streaming

✓ 4.4.5 Display

# Communication Subsystem Verification Plan

| Requirement | Status | Remaining Action | Target |
|---|---|---|---|
| 5.3.1 Survivability | ◐ Comms Pass (0% loss calm) | Test with wave / stormy conditions on lake | Wk 9 |
| 5.4.3 Data Relay | ◐ Partial (Position ✓) | Integrate Jamie's sensor pipeline; demo full telemetry stream | Wk 11 |
| 4.4.1 Autonomy & Control | ◐ Partial (commands ✓) | Implement formation control (Step 6); pool demo all 4 formation types | Wk 13-14 |
| 5.4.2 Inter-Drone Comms | ✓ ESP-NOW verified | Validate 100m range in pool environment | Pool Test |
| 5.4.6 Swarm Control | ◐ ESP-NOW layer done | Bridge ESP32→ Jetson via MAVLink/serial; implement MQTT pub/sub ≥1 Hz, verify <1s latency | Wk 12 |

# Propulsion Subsystem Verification Plan

- 4.1.3 - Stability
  - Drone will be placed in the water in different positions and shown to return to its running orientation
- 4.1.4 - Speed
  - Drone will be run at speed at constant power over a  measured distance in calm water, and timed.
  - This test will be conducted 3 times to determine speed
- 4.3.3 - Operating Temperature
  - Drone will be run at determined cruise power for 30 minutes and the internal temperature will be measured

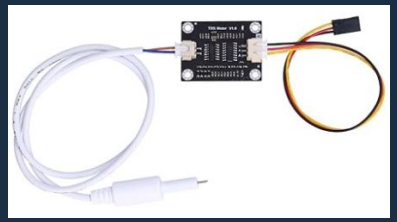
# Sensors Verification Plan




pH Probe



Temperature Probe



Conductivity Probe



GPS Module

CSV File in Format:
Date,Time, Lat., Long., Speed, GPS Fix, Temp., Cond., pH.

In order to verify full functionality of sensor suite, sensors will be tested using known, calibrated substances of varying temperature, salinity, and pH. These values will be confirmed at the ground station GUI, and necessary calibration curves will be constructed and applied.

# Sensor Requirements Verification

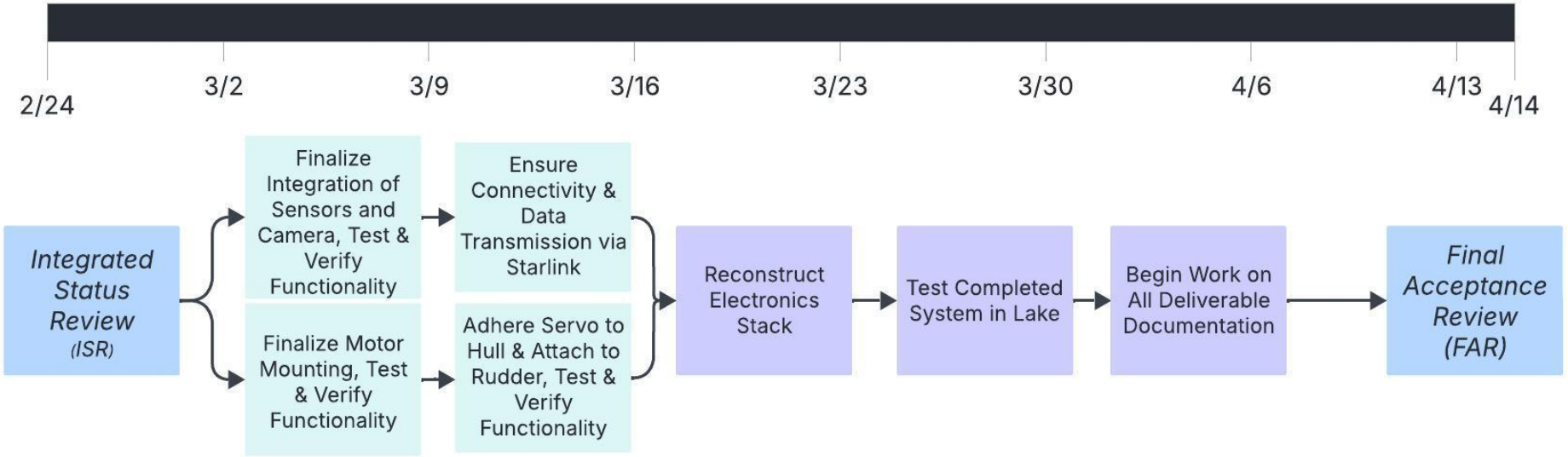
**5.1.6 Data Collection:** Each AQUABOT drone shall measure temperature, salinity, pH, and capture visual imagery at a minimum of 1080p resolution, 30 fps.

**5.4.5 Display:** The display shall show half the display with Geo-overlay and the other half will display live-streaming.

**5.2.4 Payload Mounts:** The hull shall provide standardized, sealed mounts for Starlink terminal, battery, and sensor payload.

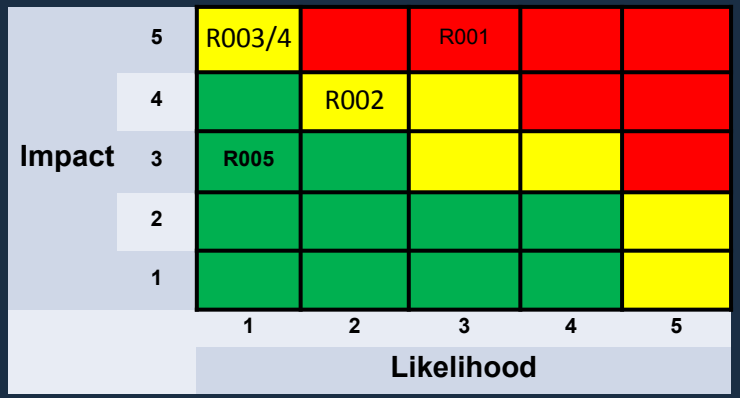
**5.3.2 Waterproofing:** Each aquatic AQUABOT drone shall meet IP68 ingress protection standards IAW IEC 60529:1989 + AMD1:1999 CSV. Waterproof components shall be tested in 1.5 m depths to ensure compliance.

# Final Acceptance Test Plan



# Risk Analysis

# Risk Analysis: Integration




| | |
|---|---|
| | High Risk |
| | Moderate Risk |
| | Low Risk |

| Impact | Likelihood |
|---|---|
| 5 - Catastrophic | 5 - Near Certainty |
| 4 - Major | 4 - Highly Likely |
| 3 - Moderate | 3 - Likely |
| 2 - Minor | 2 - Somewhat Likely |
| 1 - Insignificant | 1 - Not Likely |

| ID | Risk Description | Severity (L x I) |
|---|---|---|
| R001 | GPS accuracy leads to swarm control problems | 15 |
| R002 | Structural motor mount failure | 8 |
| R003 | Battery failure via thermal runaway | 5 |
| R004 | Bus bars unable to mitigate voltage drop across components | 5 |
| R005 | Jetson unable to establish connection to Cloud | 3 |

# Risk Analysis: Mitigation Plan

R001 (High): Swarm Control (ESP32) to Flight Controller (PIXHAWK) connection

- Test control code through simulated environment (ideal conditions)
- Initial/ final environment → low res GPS data (5m accuracy)
  - Determine if a secondary system is needed for maneuvers before separation
- Separation distance set to 50% of max range

R002 (Moderate): Structural failure of motor mount/ flange

- Conduct structural tests on ABS-CF motor mount and flange at max RPM for 1 hour
  - Determine if ABS-CF is enough in real environment → if not: use machine shop SS
  - Determine if shaft is thick enough

# Risk Analysis: Mitigation Plan

R003 (Moderate): Battery failure

- Safely charge battery when in use
- Safely discharge batteries when not in use
- Use BMS and kill switch to shut off battery if failure does happen

R004 (Moderate): Voltage drop across bus bars

- ‘Physical’ Mitigations:
  - Use overrated bus bars at 250A, when ~25A is needed
  - Use separate battery for motor (largest instant voltage drop)
  - Minimize wire distance for high current components
- Conduct testing at max electrical load to ensure no severe voltage drops

# Cost Analysis and Deliverables

# Cost Analysis

| Available Balance | $2,580.39 |
|---|---|
| Transactions Total | $ 1,819.61 |

| Date | Vendor | Status | Description | Order Number | Total | Ordered Initials | Request Number | Use Tax | Date Reconciled | Reconciled Initials | eDoc # | Notes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | EXPENSE TOTALS | | $ 1,819.61 | | | $ - | | | | |
| 10/30/2025 | Amazon | Arrived | Jetson Orin Nano Super Developer Kit | 114-0812509-67 | $ 270.66 | PK | PRF26-268 | | 11/7/2025 | NA | 32592351 | |
| 10/30/2025 | Amazon | Arrived | GoPro Camera with Touch Screen | 114-3761880-54 | $ 255.45 | PK | PRF26-269 | | 11/7/2025 | NA | 32592354 | |
| 11/13/2025 | Amazon | Return | Returning GoPro camera (114-3761880-5410649) | Return | $ (249.02) | NH | see line 19 | | 12/8/2025 | NA | 32820933 | Amazon withheld |
| 11/13/2025 | Starlink | Arrived | 1-month Roam 50GB service (11/12-12/12) | INV-DF-USA-58 | $ 48.42 | NH | PRF26-474 | | 11/18/2025 | NA | 32709737 | |
| 11/18/2025 | Amazon | Arrived | Go Pro Hero7 White | 112-1797575-20 | $ 255.45 | PK | PRF26-472 | | 11/26/2025 | NA | 32759665 | |
| 11/18/2025 | Amazon | Arrived | Internal Solid State Drive | 112-5429887-22 | $ 21.73 | PK | PRF26-472 | | 11/26/2025 | NA | 32759676 | |
| 11/19/2025 | Amazon | Arrived | Flight Control Modules | 112-5128183-51 | $ 10.86 | PK | PRF26-494 | | 11/26/2025 | NA | 32773239 | |
| 11/26/2025 | Amazon | Arrived | Lipo battery charger, lipo bag | 111-7946268-80 | $ 159.55 | DO | PRF26-559 | | 12/4/2025 | NA | 32845279 | |
| 12/3/2025 | APISQueen aka | Arrived | Brushless motor, 80A ESC | 242960 | $ 181.40 | NH | PRF26-617 | | 12/9/2025 | NA | 32884059 | ETA 12/19 |
| 12/9/2025 | Amazon | Return | Returning GoPro camera (112-1797575-2021067) | Return | $ (249.02) | NH | see line 22 | | 1/6/26 | PK | 33033272 | Amazon withheld |
| 12/12/2025 | Starlink | Arrived | 1-month Roam 50GB service (12/12-1/12) | INV-DF-USA-61 | $ 50.00 | NH | PRF26-474 | | 12/18/25 | PK | 32951831 | |
| 1/16/2026 | 4imprint | Polos | Team Polo Shirts | | $ 119.39 | PK | Debbie/PO | | | | | |
| 1/12/2026 | Starlink | Arrived | 1-month Roam 50GB service (1/12-2/12) | INV-DF-USA-65 | $ 50.00 | NA | PRF26-474 | | 1/22/2026 | NA | 33136054 | |
| 1/26/2026 | Siraya Tech | Arrived | Fibreheart ABS-CF Filament | ST60703 | $ 28.27 | PK | PRF26-738 | | 2/2/2026 | NA | 33256539 | |
| 1/27/2026 | Creality | Arrived | (Live Exclusive)Hyper Series ABS 3D Printing Filament 1kg | TK7J700HZ | $ 24.62 | NA | PRF26-739 | | 2/3/2026 | NA | 33268705 | |
| 1/27/2026 | Amazon | Arrived | Nossle | 114-6753096-48 | $ 27.07 | NA | PRF26-736 | | 2/5/2026 | NA | 33278367 | |
| 1/27/2026 | Amazon | Arrived | Spray | 114-6753096-48 | $ 32.33 | NA | PRF26-736 | | 2/5/2026 | NA | 33320701 | |
| 1/29/2026 | Amazon | Arrived | 10pcs terminal connectors, 2pcs male to female connector adapter, 16 g | 112-1318885-29 | $ 75.39 | NH | PRF26-887 | | 2/10/2026 | NA | 33280458 | |
| 1/29/2026 | Amazon | Arrived | 10pcs terminal connectors, 2pcs male to female connector adapter, 16 g | 112-1318885-29 | $ 20.31 | NH | PRF26-887 | | 2/10/2026 | NA | 33308099 | |
| 2/10/2026 | Amazon | Ordered | Railway Enclosed DC Converter | 113-6660834-61 | $ 55.09 | PK | PRF26-1034 | | 2/18/2026 | NA | 33408521 | |
| 2/10/2026 | Amazon | Arrived | Pixhawk Flight Controller , Starlink 12V Power Supply | 113-6551329-21 | $ 219.55 | PK | PRF26-1034 | | 2/19/2026 | NA | 33421766 | |
| 2/10/2026 | Amazon | Ordered | Waterproof Analog Servo | 113-1658434-05 | $ 52.72 | PK | PRF26-1034 | | 2/19/2026 | NA | 33421769 | ETA 2/17-2/19 |
| 2/11/2026 | Amazon | Arrived | Sealant and Coaxial | 113-0787254-02 | $ 38.94 | NA | PRF26-1060 | | 2/19/2026 | NA | 33423541 | |
| 2/11/2026 | Amazon | Arrived | M4 Screws and Wrenchs | 113-1184687-34 | $ 24.40 | NA | PRF26-1057 | | 33423545 | NA | 33423545 | |
| 2/11/2026 | SaltWater | Arrived | 3 x PINPOINT pH Probe Lab Grade Double Junction | 100796297 | $ 130.41 | NA | PRF26-1059 | | 2/17/2026 | NA | 33422040 | |
| 2/12/2026 | Atlas Scientific | Arrived | EZO™ pH Circuit | 447439 | $ 155.87 | NA | PRF26-1061 | | 2/20/2026 | NA | 33454357 | |
| 2/20/2026 | Amazon | Ordered | 1010 O rings Kit | 113-0771434-4 | $ 9.77 | NA | PRF26-1128 | | | | | |

Within Budget

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# Deliverables

| Deliverable | Date | Description |
|---|---|---|
| Programmed Cameras and GUI | 12/9/2025 | Camera connected and linked to GUI |
| Programmed Sensor Suite | 12/9/2025 | Sensors and GPS integrated |
| Materials CAD | 12/9/2025 | CAD for all major systems complete |
| Communication Module | 2/26/2026 | Starlink integrated with all components & GS. |
| Propulsion System | 3/5/2026 | Functional propulsion system w/ motor control |
| Operational Ground Station | 2/28/2026 | GUI and backend for live data transmission |
| Functional Hull Prototype | 3/5/2026 | Hull integrated with all components |
| Manual | 4/14/2026 | Includes documentation and user guides |
| Project Final Report | 5/5/2026 | Includes summary, system specs, and results |

# Summary

# Summary

Major Accomplishments:

- Via thorough testing, several requirements have already been fully passed (inter-drone communication, ground station communication, etc.)
- Communication between two drones with preparations for communication between several drones has been completed
- Significant progress towards physical integration of motors and sensors has been made using redesigned mounting systems
- Significant progress has been made towards finalizing the GUI and transmitting data to the ground station

Moving towards FAR:

- Most concerned about inconsistencies in the environment affecting the system
  - Waves
  - Weather
  - Network/radio traffic

# Questions & Discussion

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